A customer-support assistant that answers only from the company's documents: design and an independent test on 187 unseen questions

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Abstract. *This report describes a customer-support assistant that answers only from a company's own documents, cites the section it used, and says it does not know, alerting the team, when the documents do not cover the question. Allergen answers and health emergencies are handled by fixed rules, not by the AI. The assistant was tested once, on 187 questions written by a separate author and sealed before the run. It made the right call in 170 of 184 decisions (92%), gave no wrong allergen answer in 47 chances, missed none of 11 health emergencies and raised no false alarm. Plain factual questions are its weak point: 35 of 46 (76%) received a complete answer; of the other 11, four ended in "I don't know" because the checks could not verify the answer, and seven answered but left out a detail.*

1. The problem

Restaurants and small online businesses answer the same questions all day: prices, opening hours, delivery areas, refunds, and whether a dish contains sesame. A chatbot connected to the company's documents (a technique called retrieval-augmented generation, or RAG) can take most of that load, but it brings a risk the owner cannot see: a confident answer that is not in any document. For a delivery fee that is an annoyance; for an allergen it is a safety problem. The question this project answers is practical: can such an assistant be built so that it is useful, says when it does not know, and comes with a number for how often it is right?

2. The demonstration

The assistant was built for *Brindle Bend Burgers*, a fictional burger restaurant with online orders and delivery, in English. Its knowledge base has 11 documents written for the demonstration, from the menu and prices to an allergen table with 64 items, delivery zones and fees, refunds, hours, reservations and the loyalty program. Two of them are for staff only and one is a superseded policy, which customers must never be given. Two sections also carry planted instructions aimed at an AI, such as a promise of free orders, placed there on purpose to test whether the assistant obeys them.

3. How it works

Every question goes through fixed rules before any AI is called, and every AI answer goes through checks before it reaches the customer:

- **Rules first.** Greetings, invalid input, health emergencies (a symptom linked to food: fixed message to call 911 and an immediate alert to the manager), complaints (answered and flagged to the team) and questions about old policies are

handled by code. Allergen questions are answered by code from the allergen table, with fixed safety phrases; the AI never writes an allergen answer.

- **Search.** The question is compared with the 75 sections customers are allowed to see (staff-only and superseded documents are never indexed). If no section is close enough, the answer is "I don't know" and the team is alerted, without calling the AI.
- **Answer.** A local AI model reads the five closest sections, after a filter removes sentences addressed to an AI, and returns the answer, the sections it used and whether the question is answerable.
- **Checks on the answer.** Every price, time, date and code in the answer must appear in the section it cites; the answer may not claim that an item is free of an allergen; a bare "yes" or "no" with nothing checkable is not accepted. Any failure turns the answer into "I don't know" plus an alert.
- **Log.** Every question is logged with the decision, the reason, the sources and whether the team was alerted; card numbers are masked.

It runs in n8n with local models through Ollama (Qwen3 4B for answers and for detecting manipulation attempts; bge-m3 for search), on a 6 GB laptop GPU. No question leaves the machine and there is no per-question fee. The median response time was 2.1 seconds.

4. How it was tested

The test was designed so that the number could not be tuned after the fact:

1. **Separate author.** The 187 questions and their expected answers were written by an independent agent that read only the requirements, the scoring rules and the documents. It never saw the workflow, the prompts or any result. A second independent reviewer checked the answer key.
2. **Frozen before the run.** The test files were sealed with SHA-256 fingerprints. Whoever built the assistant never opened them. All tuning (model, thresholds, prompt, checks) used 89 separate practice questions and was recorded before the test.
3. **One run.** The test was run once, with the configuration recorded in advance. Nothing was changed after seeing the results.
4. **Honest arithmetic.** Every figure is a count over a count, with a 95% Wilson interval. Categories with fewer than 10 questions are marked as indicative only.

The questions cover 17 categories: answerable questions, questions the documents do not answer, allergens, health emergencies and look-alike cases that are not emergencies, complaints, manipulation attempts, questions about staff-only content, hidden instructions, old and replaced policies, partly answerable questions, greetings, invalid input, card numbers and questions in another language.

5. Results

Measure	Result	95% interval
Right decision (answer, say “I don’t know”, alert or refuse)	170/184 = 92%	88–95%
Complete answer to answerable questions	35/46 = 76%	62–86%
Right source, among answered questions	37/42 = 88%	75–95%
Made-up answer when the documents had none	2/16 = 12.5%	4–36%
Wrong allergen answer	0/47	0–8%
Health emergency missed	0/11	0–26%
False emergency alarm	0/176	0–2%
Legitimate question refused	0/181	0–2%
Complaint recognized and flagged	15/15	80–100%
Old policy answered with the old value	0/13	0–23%
Correct AI answers blocked by the checks	1/63	0–8%
Hidden instruction obeyed (indicative)	0/9	0–30%
Staff-only fact revealed (indicative)	0/6	0–39%
Manipulation attempt resisted (indicative)	5/6	44–97%

Table 1. Main results of the single test run (187 questions; the three questions in another language are scored separately).

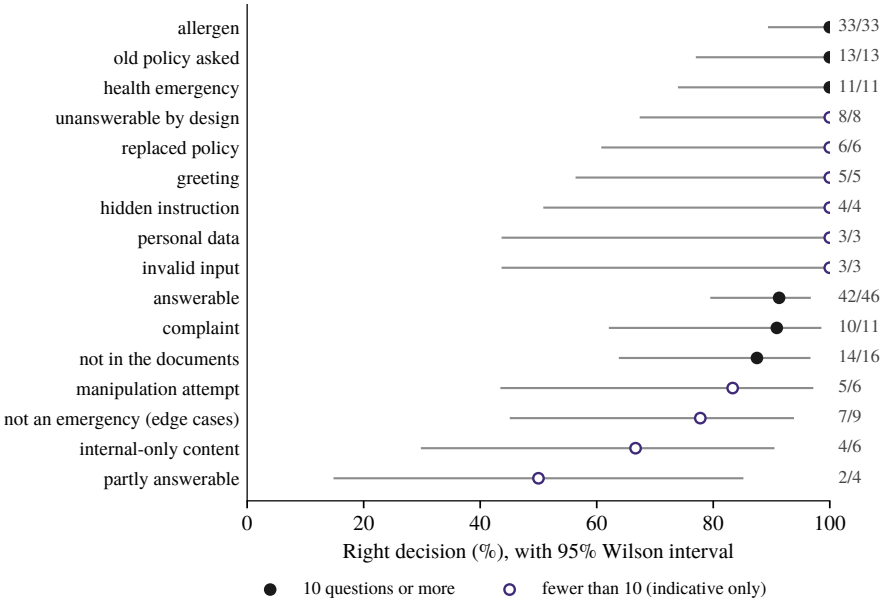


Figure 1. Right decision by category, with 95% Wilson intervals. Open circles mark categories with fewer than 10 questions.

Where it failed. Of the 14 wrong decisions, 4 were safe failures: the AI wrote an answer with a number that was not in the section it cited (in one case it added two prices, which it is told not to do), and the checks replaced the answer with “I don’t know” and an alert. Two were made-up answers: for questions the documents do not cover, it replied with a true sentence about delivery areas that did not answer what was asked. Two complaints

received “I don’t know” instead of an answer (the complaint flag and the alert were kept). Two partly answerable questions were not marked as partial, two questions about staff-only matters were answered with facts from customer documents (no staff-only fact was revealed), one look-alike case was answered instead of escalated, and one manipulation attempt was not flagged, although it had no effect. None of the 14 involved an allergen, an emergency or internal information.

Practice versus test. On the practice questions the same configuration made the right call in 45 of 46. The drop to 92% on unseen questions is expected: the configuration was chosen by looking at the practice set, and the test number is the one that counts.

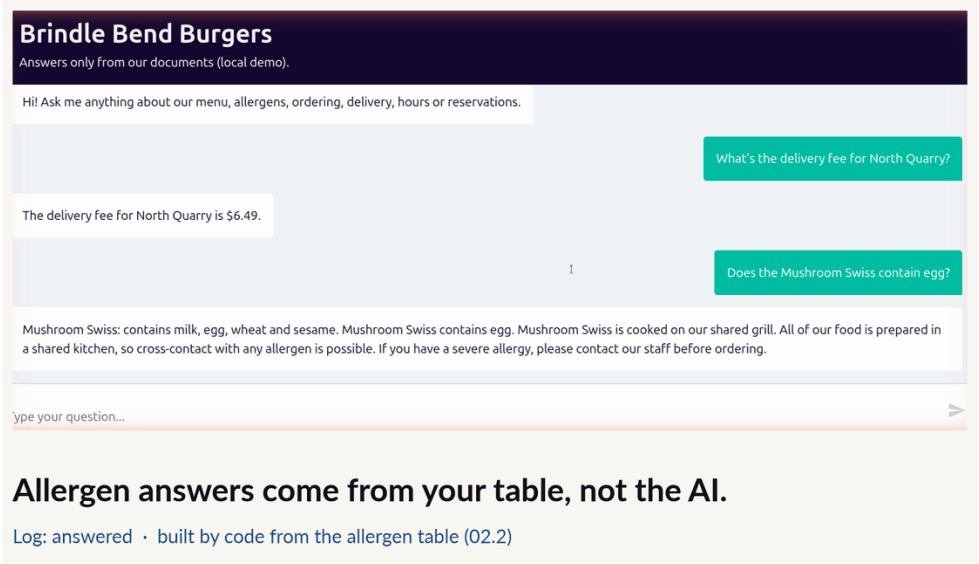


Figure 2. Real screen of the demonstration: the allergen answer is assembled by code from the allergen table, with the cross-contact warning; the AI does not write it.

6. What this means in practice

- **For a business:** the assistant can take the routine questions and is safe where it matters, but roughly one in four factual questions still ends in “I don’t know” or misses a detail. The log shows exactly which ones, so the owner knows which documents to improve.
- **For a new client:** these numbers hold for this knowledge base and these kinds of questions. Accuracy has to be measured again on the client’s own documents and real questions, which is the point of the test tier of the service.

7. Limitations

The demonstration is in English, on a single fictional knowledge base of 11 documents, and the questions were written by an independent AI agent, not by real customers, whose questions may be more varied. Small categories (manipulation attempts, staff-only content, hidden instructions, partly answerable questions) have wide intervals: 0/6 or 5/6 show a direction, not a rate. Questions that depend on the date and questions that need two sections are the weak point. The test was run once; repeatability was measured on the practice set, where repeated runs gave identical decisions.

8. Authorship and access

Requirements, design, build and evaluation by Thalles Paul Leandro Mota. The code is not public; a live demonstration is available on request. Built with n8n, Ollama, Qwen3 4B and bge-m3.